

Consolidated progress report: 2D nucleus segmentation after the Scientific Reports rejection

29 September 2026. This document consolidates everything from the rejection through the present experiments: what the reviewers found, what our own audit found afterwards, what we re-ran, and what the new public-data experiments measure. All numbers are read from committed result files. Companion documents: `prediction_samples.pdf` (qualitative evidence), `changes_log.pdf` (itemised changes), `code_audit.pdf`, `reviewer_action_matrix.pdf`, `next_steps.pdf`.

Scope. The task is 2D semantic nuclear foreground segmentation. This is not whole-cell segmentation, not instance separation, not cancer diagnosis, and not a clinically validated system. No claim in this document depends on a result we have not measured.

Part I — Where we started: the rejection and what it was right about

Scientific Reports rejected submission a69523d8 on the editor's judgement that Reviewer 1 had identified "fundamental issues that cannot be addressed", with Reviewer 2 adding four concerns (insufficient ablations, missing statistics, thin dataset description, too few baselines). The submitted manuscript claimed 95.1% Dice and 89.2% IoU from 599 training annotations, against seven baselines scoring 46.6–75.5%.

We treated the review as correct and audited our own material rather than rebutting. Five findings emerged, all verified against the code, the released checkpoint and the data on disk.

1. The original image-wise split has strong leakage indicators. Fifty percent of test images had a near-duplicate in training (normalised cross-correlation > 0.95 ; median maximum correlation 0.950). Because source-volume metadata are unavailable, we conservatively reconstructed likely source groups from frame geometry and inter-image correlation. Keeping those inferred groups disjoint drops the maximum cross-split correlation to 0.61, but does not prove biological independence.

2. The reported headline number does not match its own per-image data. The stored summary for the proposed model records mean Dice 0.9513145 and mean IoU 0.9024580. Recomputing from the 128 saved per-image values in the same directory gives 0.9413145 and 0.8924580 — each stored mean is higher by exactly 0.01. The comparison CSV agrees with the recomputed values. This is an arithmetic finding about stored files; it does not establish how the number was entered, and we adopt neither the inflated nor the recomputed value as a validated result.

3. "60% of images are empty" was an encoding artefact. 508 of the masks are stored as 0/1 and 347 as 0/255. The statistics script thresholded every mask at 127, so the 0/1 half read as empty. All 855 valid images contain nuclei (mean foreground 8.3%); two further files contained image data saved as labels and were excluded. The same threshold appeared in the sampler, which therefore down-weighted 508 nucleus-containing images by 8.9x — the "complexity-weighted sampling" component did not do what the manuscript described.

4. The published algorithm is not the trained objective. The released checkpoint was trained with $0.7 \cdot \text{Dice} + 0.3 \cdot \text{adaptive-weighted BCE}$. It contains no Tversky term and no boundary term, both of which the manuscript's Algorithm 1 specified. Reviewer 1's comments 4 and 5 follow directly from this.

5. Implementation defects. Gabor kernels were normalised by their signed sum, which is near zero for 8 of the 24 kernels; validation discarded low-scoring batches through an IQR filter before checkpoint selection; loaders silently substituted zeros for unreadable files; and 16-bit images were loaded through an 8-bit conversion that kept only the high byte.

Part II — Re-running the private study correctly

We rebuilt the private-data evaluation from the split upward: conservative inferred-source-group partitioning, one identical 30-epoch budget for every architecture, corrected mask decoding, L1-normalised Gabor kernels, native-resolution metrics, and three seeds for the proposed model. Thirty-eight runs completed.

The crucial correction is statistical. Images are clustered within inferred source groups, so per-image bootstrap confidence intervals overstate precision. Recomputing with the inferred group as the resampling unit (7 groups) widens every interval to the point where the differences disappear.

Condition (private data, inferred-group split)	Dice	95% CI (cluster bootstrap, inferred-group level)	Groups
w/o Gabor channel	0.9290	[0.9083, 0.9487]	7
w/o SCSE	0.9329	[0.9135, 0.9502]	7
w/o weighted sampling	0.9277	[0.9087, 0.9488]	7
plain Dice+BCE loss	0.9366	[0.9199, 0.9529]	7
plain U-Net decoder	0.9255	[0.9062, 0.9475]	7
U-Net ResNet-50 baseline	0.9320	[0.9140, 0.9502]	7
Attention U-Net baseline	0.9302	[0.9127, 0.9486]	7

Every interval overlaps every other. Under a correct analysis the private data cannot distinguish the proposed configuration from a plain U-Net baseline, and the ablations that the original manuscript reported as catastrophic (−80 to −95 Dice points) are not reproducible: the largest single-component effect we measure is under one Dice point.

Four-fold leave-fields-out cross-validation over all 23 fields gives pooled Dice 0.894, below the single hold-out estimate of 0.933, and exposes the reason: the two annotation batches use different boundary conventions. Fields whose masks are stored 0/255 yield recall ≥ 0.98 with precision 0.72–0.89 (reference drawn tighter than the model); several 0/1 fields invert this, precision ≥ 0.96 with recall 0.73–0.83 (reference drawn wider). Cross-convention testing costs 0.05–0.10 Dice for reasons unrelated to segmentation quality.

The unresolved problem. Our field grouping is derived from frame geometry and inter-slice correlation, not from acquisition metadata. It is a reasonable reconstruction, but it is not proof of biological independence. Until source-volume identifiers are recovered, no private-data result should be presented as leakage-proof.

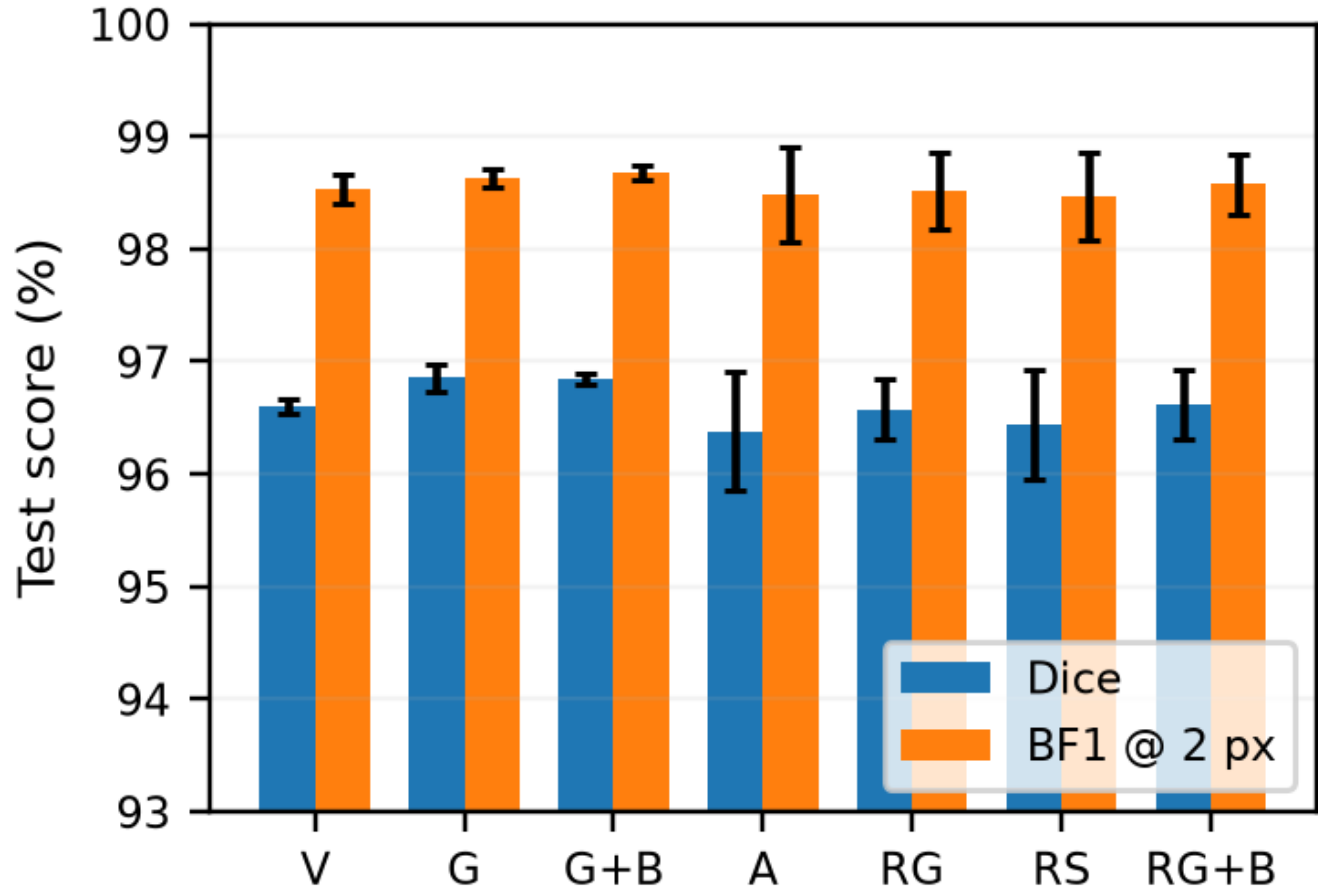
Part III — New matched public-data experiments

Because the private provenance cannot currently be certified, we moved the controlled experiments to public data with official splits: **BBBC039** (U2OS fluorescence nuclei, CC0) for training and in-domain testing, and **TNBC v1.1** (triple-negative breast histology, CC BY 4.0) as an untuned transfer stress test.

Protocol, fixed before the runs and hashed: 100 training fields, 50 validation, 50 test; 20 epochs; AdamW; identical augmentation; uniform sampling for every method; fixed threshold 0.5; no test-time augmentation; checkpoint chosen on unsmoothed full-validation Dice; test partition touched only afterwards. 21 runs, 420 epochs, three seeds per configuration.

Configuration	BBBC039 Dice [95% CI]	seed SD	BF1@2px	HD95 px	Δ vs vanilla	p (Holm)	TNBC patient Dice
Vanilla B7	0.9660 [0.9640, 0.9679]	0.0007	0.9853	1.324	—	—	0.0807
Gabor bundle	0.9685 [0.9667, 0.9703]	0.0013	0.9863	1.278	+0.0025	<0.001	0.0294
Gabor + boundary	0.9684 [0.9666, 0.9701]	0.0005	0.9868	1.275	+0.0023	<0.001	0.0386
SCSE-only B7	0.9638 [0.9617, 0.9657]	0.0053	0.9848	1.349	-0.0022	<0.001	0.0436
Residual Gabor	0.9657 [0.9636, 0.9677]	0.0027	0.9852	1.306	-0.0003	0.965	0.0404

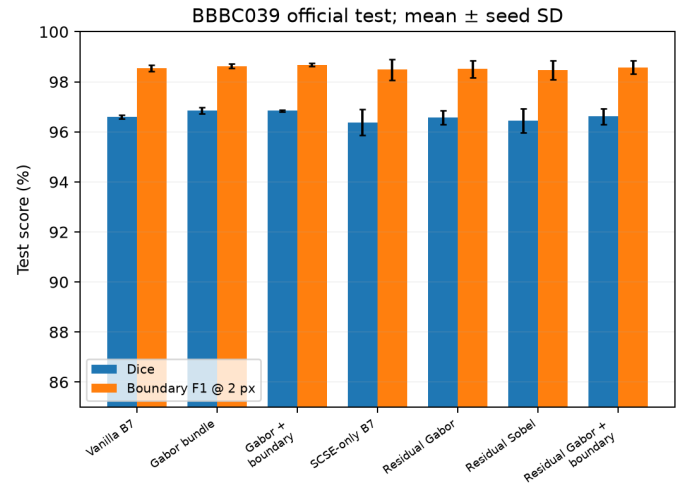
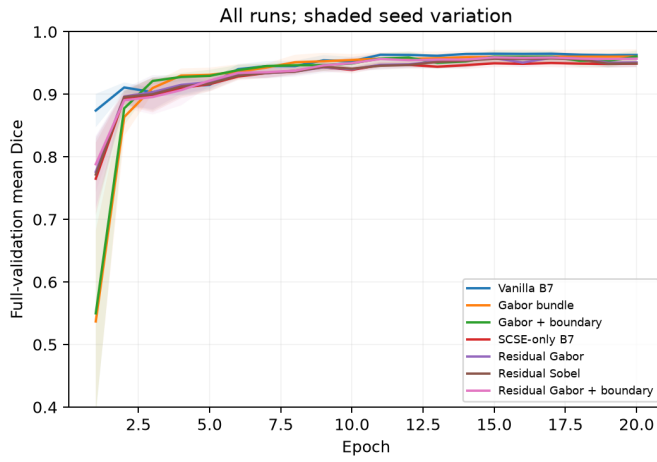
Configuration	BBBC039 Dice [95% CI]	seed SD	BF1@2px	HD95 px	Δ vs vanilla	p (Holm)	TNBC patient Dice
Residual Sobel	0.9644 [0.9623, 0.9664]	0.0048	0.9847	1.348	-0.0016	<0.001	0.0443
Residual Gabor + boundary	0.9661 [0.9640, 0.9681]	0.0031	0.9858	1.286	+0.0001	0.965	0.0449



In-domain, the edge bundle gives a real but negligible gain. Gabor + projection + SCSE improves on vanilla B7 by 0.250 Dice points ([0.197, 0.299], Holm-corrected $p < 0.001$). The effect is statistically clear and practically immaterial: it is a quarter of one Dice point on a benchmark where every configuration scores above 0.963.

Most of the bundle does not help. SCSE alone is worse than vanilla (0.9638, -0.0022). Residual Sobel fusion is worse (0.9644). Residual Gabor fusion is indistinguishable from vanilla (0.9657, $p = 0.965$). Adding the explicit boundary term to the Gabor bundle does not change Dice (0.9684) but does improve boundary F1 measurably, which is the one place the term behaves as designed.

Out of domain, every added component hurts. Transferred to breast histology without tuning, vanilla B7 reaches patient-level Dice 0.0807 and every edge-enhanced variant scores lower (0.0294-0.0449), all differences significant after correction. All values are near zero: fluorescence-to-H&E transfer fails outright. The qualitative failure is shown in prediction_samples.pdf.



Validation-only Gabor sensitivity and empty-image correction

We then froze a separate 30-run, 600-epoch sensitivity study using only the official 100-image training and 50-image validation partitions; no BBBC039 test case was decoded or evaluated. An initial two-run pilot revealed that the validation partition contains 49 foreground images and one empty image. Correct prediction of that one case can move the all-case mean by about two Dice points, so the pilot was preserved but excluded. The full study selected checkpoints on foreground-only Dice and reports the empty case separately.

Condition	Foreground Dice	seed SD	foreground BF1@2px	Dice delta, pp	paired bootstrap interval, pp	p (Holm)
Gabor reference (8 orientations, 3 scales)	0.966840	0.001701	0.982265	—	—	—
4 orientations	0.966858	0.001418	0.982505	+0.0018	[-0.0086, +0.0121]	0.723
12 orientations	0.966642	0.001500	0.982285	-0.0198	[-0.0360, -0.0033]	0.010
1 scale (0.15 cycles/px)	0.967229	0.000713	0.982660	+0.0389	[-0.0097, +0.0777]	<0.001
5 scales	0.967147	0.001519	0.982839	+0.0307	[+0.0138, +0.0537]	<0.001
sigma 3	0.965405	0.004901	0.981525	-0.1435	[-0.1713, -0.1161]	<0.001
sigma 7	0.967091	0.001018	0.983049	+0.0250	[+0.0098, +0.0408]	0.010
Sobel control	0.967272	0.000690	0.983639	+0.0431	[-0.0022, +0.0949]	0.182
Neutral-edge control	0.965864	0.000506	0.981424	-0.0977	[-0.1480, -0.0602]	<0.001
Weighted sampling	0.967313	0.000674	0.983271	+0.0473	[+0.0218, +0.0795]	<0.001

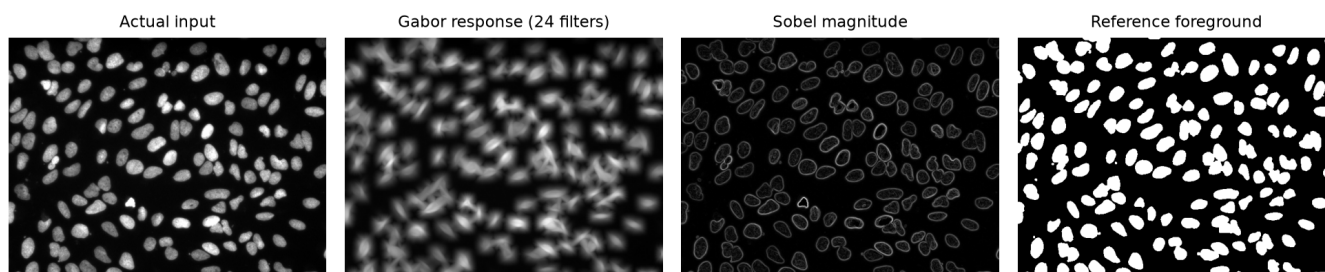
The strongest reproducible changes are very small. Five Gabor scales improve foreground Dice by 0.0307 percentage points ([+0.0138, +0.0537]); sigma 7 by 0.0250 points ([+0.0098, +0.0408]); and weighted sampling by 0.0473 points ([+0.0218, +0.0795]). One-scale Gabor and Sobel have positive means but bootstrap intervals that cross zero. The neutral-edge control is 0.0977 points below the reference, supporting a small benefit from edge information within this architecture, but not a uniquely optimal 24-filter Gabor

design. Sigma 3 is unstable across seeds and 0.1435 points worse. The single empty case is descriptive only: one-scale and sigma-7 models predict it empty in all three seeds, whereas weighted sampling does so in only one seed. These validation results are exploratory because BBBC039 had already informed development and cannot replace confirmation on a new acquisition.

Part IV — Historical checkpoints re-evaluated independently

All 24 supplied historical checkpoints from both export versions were re-evaluated on public data under one protocol (72 evaluations). On BBBC038 they score 0.46–0.71 Dice; on BBBC039 the best reaches 0.56 and several collapse to 0.00. These are not matched-training comparisons — the checkpoints were trained on different data with different pipelines — so they establish only that the historical models do not transfer, not that any architecture is superior. No run was discarded for scoring poorly; the failures are retained in the summary.

Part V — The Gabor channel, shown honestly



This figure replaces the illustration in the rejected manuscript, which was generated with a language model and labelled "Quantum". The operation is ordinary Gabor filtering: 24 kernels over 8 orientations and 3 wavelengths (20, 6.67 and 4 px), L1-normalised, combined by per-pixel maximum response. Every panel is computed by the released code on a real test image. The word "quantum" appears nowhere in the new implementation, and no quantum-mechanical mechanism is claimed or implied.

Part VI — What we can and cannot claim

Supported by measurement. Nuclear foreground segmentation on fluorescence data reaches 0.963–0.969 Dice with boundary F1 above 0.984 and HD95 near 1.3 px, under a locked protocol with official splits, equal budgets, three seeds and corrected statistics. The edge bundle gives a statistically detectable in-domain gain of about a quarter of a Dice point and a consistent out-of-domain penalty. The private confocal models produce accurate masks whose residual error is dominated by annotation-convention offsets rather than detection failures.

Not supported. Any superiority claim over modern matched-data baselines; any statement about breast histology, cancer detection or clinical use; instance separation; leakage-proof private-data validation; and the 95.1% figure in the rejected manuscript, which we do not carry forward in any form.

The honest position for a resubmission. The measured effect of the method's distinctive component is too small in-domain and negative out-of-domain to support a superiority paper. Two defensible directions remain: a transparent evaluation-and-audit contribution documenting how split leakage, unequal budgets and per-image statistics inflate small-data segmentation results; or a new methods contribution built on a component that demonstrably works, which we do not currently have. Resolving the private-data provenance and running the confirmatory protocol described in `next_steps.pdf` determines which is available. ISBI 2027 closes 26 October 2026.